

Artificial Intelligence (AI2002)

Final Exam

Date: 26/05/2025

Total Time: 3 hr

Course Instructors

Total Marks: 50

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Total Questions: 5

Roll No

Section

Student Signature

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ATTEMPT ALL QUESTIONS AND THEIR PARTS IN SEQUENCE

CLO1: Summarize the notions of rational behavior and intelligent agents.

Q1. Give short answers to the following in two lines:

[Marks 10, 30 min]

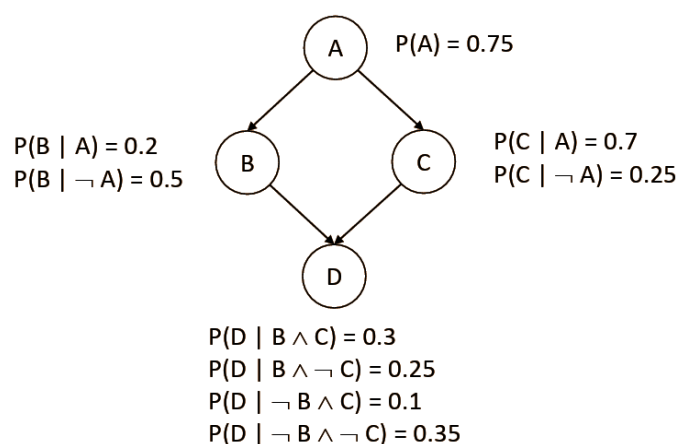
- What are the fundamental components essential for an intelligent agent to function?
- What is an actuator in an intelligent agent?
- What is Rationality in the context of an intelligent agent?
- What is the primary role of an AI problem-solving agent?
- Why is careful problem formulation crucial before applying search algorithms?
- Consider a taxi-driving scenario, where the traffic conditions and actions of other drivers are unpredictable. List all the environment types this scenario belongs to with reasons.
- How does a heuristic function improve a search algorithm compared to uninformed search?
- What is the difference between an admissible and inadmissible heuristic?
- Discuss how alpha-beta pruning improves the efficiency of Minimax Search algorithm?
- Explain the role of mutation and crossover in an evolutionary algorithm.

CLO4: Summarize concepts and approaches in knowledge representation, planning, learning, robotics and other AI areas.

Q2. Probability and Naive Bayes

[Marks 10 = 5+5, 30 min]

Part A. Consider the following Bayesian network. A, B, C, and D are Boolean random variables. If we know that A is true, what is the probability of D being true?



A financial institution is developing a system to predict whether a person will default on a loan based on the following categorical attributes:

Training data

Employed	Yes	Yes	No	Yes	No	No	Yes	Yes	No	No
Credit History	Good	Good	Bad	Bad	Good	Bad	Bad	Good	Good	Bad
Owns House	Yes	No	No	Yes	No	No	No	Yes	Yes	Yes
Default	No	No	Yes	No	Yes	Yes	No	No	No	Yes

Part B. Apply Naïve Bayes classifier to predict whether a new applicant with the following profile will default

- Employed: **No**

- Credit History: **Bad**

- Owns House: **No**

CLO4: Summarize concepts and approaches in knowledge representation, planning, learning, robotics and other AI areas.

Q3. Supervised Learning

[10 marks, 5+5, 45 min]

Part A. A mobile food delivery startup wants to predict its monthly number of customer orders based on the amount of money spent on social media advertising. The company has collected data over the past 12 months on ad expenditure and the number of customer orders as shown below:

Month	1	2	3	4	5	6	7	8	9	10	11	12
Ad Expenditure	200	400	600	800	100	700	500	900	300	1000	200	600
Customer Orders	4	7	9	11	3	10	8	12	5	14	4	9

1. Calculate the linear regression equation and interpret the coefficients.
2. Estimate the average number of customer orders for a month where \$650 is spent on advertising. Using linear regression model.

Part B. You are given the following training dataset with two features: **Exam Score** and **Project Score**.

The task is to classify a new student into either class "Pass" or "Fail" using the K-Nearest Neighbors (**KNN**) algorithm.

Student	Exam Score	Project Score	Class
A	85	70	Pass
B	80	90	Pass
C	60	60	Fail
D	70	65	Fail
E	95	85	Pass
F	55	50	Fail
G	75	70	Pass
H	50	45	Fail

A new student has **Exam Score = 72** and **Project Score = 68**.

1. For **K = 3** and **K = 5**, determine the predicted class, after finding the Euclidean distance of the new student from all the training samples.
2. Explain the effect of changing the value of K on the classification result.

CLO4: Summarize concepts and approaches in knowledge representation, planning, learning, robotics and other AI areas.

Q4. Supervised Learning

[10 marks, 5+5, 45 min]

Part A. You are given the following dataset of job applicants. The goal is to decide whether an applicant should be hired or not, using the ID3 decision tree algorithm.

Education	BS	BS	MS	PhD	PhD	BS	MS	PhD	MS	BS	MS	BS	PhD	BS
Experience	Low	Low	Med	High	High	Med	High	Med	Low	Med	Med	Low	High	Med
Skills	Poor	Good	Good	Best	Best	Poor	Good	Good	Good	Good	Best	Poor	Good	Best
Hired	No	No	Yes	Yes	Yes	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes

1. Calculate the Entropy of the full dataset for the target attribute Hired.
2. For each of the following attributes calculate the information gain:
 - Education
 - Experience
 - Skills
3. Select the best attribute as the root node of the decision tree using the ID3 algorithm.

Part B. You are working as a data scientist in a healthcare startup that has developed a machine learning model to predict whether a patient has a rare disease (Positive) or not (Negative) based on their medical records. The disease is quite rare, affecting only about 2% of the population.

The model was tested on a dataset of 1,000 patients, and it made the following predictions:

- It predicted 950 patients as Negative, and out of these, 940 were actually healthy, but 10 were sick.
- It predicted 50 patients as Positive, and out of these, 30 were healthy, but 20 were actually sick.

Perform the following:

1. Construct the Confusion Matrix based on the information provided.
2. Calculate the accuracy of the model.

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Q5. K-Means and ANN

[Marks 10, 5+5 25 min]

Part A. In a seismic monitoring system, each sensor node records its x, y location in kilometers. Your task is to cluster sensor nodes into TWO zones using the K-Means algorithm to optimize alert response regions.

You are given the following 8 sensor coordinates:

Sensor	1	2	3	4	5	6	7	8
X(km)	5	6	40	38	8	47	43	7
Y(km)	15	17	60	58	14	66	62	13

Simulate the K-Means algorithm for k = 2. Perform one iteration with data point 1 and 6 as initial cluster centers, and explain:

- Which sensors belong to which cluster.
- What will be the updated cluster centers after one iteration.

Part B. A logistics company wants to use an artificial neural network (ANN) to predict the optimal delivery category for a package based on certain attributes provided at the time of dispatch. Each package is described using the following features:

- Package weight (in kilograms)
- Dimensions (in cubic centimeters)
- Delivery distance (in kilometers)
- Package fragility score (0 to 10)
- Delivery speed preference (standard, express, overnight — encoded numerically)

The possible delivery categories are

- Standard Ground
- Express Air
- Overnight Priority
- Same-Day Urban

The company has several thousand historical delivery records and wants to automate this process.

Answer the following questions:

1. How many input values (units) will the ANN need?
2. How many output values (units) will the ANN need?
3. Should the ANN use hidden layers or not? Why?
4. What activation function is commonly used in the hidden layer?
5. Draw a simple feedforward neural network diagram for the above process.

Formulas

$$w_0 = \frac{1}{M} \sum y_i - \frac{w_1}{M} \sum x_i$$

$$w_1 = \frac{M \sum x_i y_i - \sum x_i \sum y_i}{M \sum x_i^2 - (\sum x_i)^2}$$

$$Entropy(S) = -\frac{p_{+ve}}{T} \log \frac{p_{+ve}}{T} - \frac{p_{-ve}}{T} \log \frac{p_{-ve}}{T}$$

$$Gain(S, A) = Entropy(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{S} \cdot Entropy(S_v)$$

$$Euclidean Distance (p1[x1, y1], p2[x2, y2]) = \sqrt{(x2 - x1)^2 + (y2 - y1)^2}$$

$$\mathbf{P}(Cause, Effect_1, \dots, Effect_n) = \mathbf{P}(Cause) \prod \mathbf{P}(Effect_i | Cause).$$